

Agentic AI Course - Day 10

Krish Naik - 10h course | Video 6:00-6:40

Vectorless RAG

Vectorless RAG retrieves relevant context WITHOUT a vector database - using keyword search, the LLM itself, or structured filters. Useful when data is small, changes fast, or a vector DB is overkill.

Key concepts

- Not every RAG needs embeddings - sometimes keyword/BM25 or SQL filters are enough.
- LLM-as-retriever: let the model pick relevant sections from provided text.
- Pros: simpler stack, no index to maintain, fresh data.
- Cons: can be slower/costlier per query and weaker on pure semantic matches.

Code example

```
# Example: filter docs by keyword, then let the LLM answer from them
hits = [d for d in docs if 'refund' in d.lower()]
answer = (prompt | llm | StrOutputParser()).invoke({'context': '\n'.join(hits), 'question': q})
```

Today's tasks (1.5h)

- [] Try keyword/BM25 retrieval
- [] Compare answers vs vector RAG
- [] Note when vectorless is the better choice

Resources

Video: youtu.be/rV3HJ4LEZ7k (start at 6:00-6:40)

Code: github.com/krishnaik06/RAG-Tutorials

(Full clickable links are in the app: Goals > this day > Study block.)